Homework #7 - Spaceflight Mechanics

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1 Part 1

1.1 Problem

Draw the intersection of the sphere of constant moment and the ellipsoid of constant energy and highlight vectors $\vec{H}_c$ and $\vec{\omega}$. Then do the same for the ellipsoids of constant moment and constant energy.

1.2 Initialization of values

We first need to define some values, like the maximum and minimum kinetic energies, useful to define the moments of inertia (min and max), and thus how the mass is distributed in the body (in the case of an homogeneous material body, that will also tell us its shape). We use the values suggested by the exercise, but we choose X_1 and X_2, respectively selecting the middle moment of inertia J_2 and the kinetic energy the body possesses E_r.

$$|\vec{H}_c| = 1$$

$$E_{min} = 1 \qquad \Rightarrow \qquad J_1 = 0.5 \cdot \frac{H_c}{E_{min}} = 0.5$$

$$E_{max} = 2 \qquad \Rightarrow \qquad J_3 = J_1 \cdot \frac{E_{min}}{E_{max}} = 0.25$$

$$X_1 = 0.2$$

$$E_2 = E_{min} + X_1(E_{max} - E_{min}) \qquad \Rightarrow \qquad J_2 = J_1 \cdot \frac{E_{min}}{E_2} = 0.4167$$

$$X_2 = 0.4$$

$$E_r = E_{min} + X_2(E_{max} - E_{min}) = 1.4$$

If we assume $\omega_{20} = 0$ to simplify matters, we only need to obtain ω_{10} and ω_{30}. We can do this by solving the following simplified system of equations.

$$\begin{cases} J_1^2\ \omega_{10}^2 + J_3^2\ \omega_{30}^2 = H_c = 1 \\ \frac{1}{2} J_1\ \omega_{10}^2 + \frac{1}{2} J_3\ \omega_{30}^2 = \tilde{\epsilon} \end{cases}$$

Which leads directly to

$$
\begin{aligned}
\omega_{10} &= \sqrt{\frac{1-2J_3E_r}{J_1(J_1-J_3)}} = 1.5492\ rad/s \\
\omega_{20} &= 0\ rad/s \\
\omega_{30} &= \sqrt{\frac{1-2J_1E_r}{J_3(J_3-J_1)}} = 2.5298\ rad/s
\end{aligned}
$$

$$
\begin{aligned}
&\Rightarrow\ \vec{\omega} = \begin{bmatrix}\omega_{10} & \omega_{20} & \omega_{30}\end{bmatrix} \\
&\Rightarrow\ \vec{H}_c = \begin{bmatrix}J_1\omega_{10} & J_2\omega_{20} & J_3\omega_{30}\end{bmatrix}
\end{aligned}
$$

1.3 Spheres and ellipsoids

We can then draw the sphere of constant moment, the ellipsoid of constant energy and their intersection.

$$
\begin{aligned}
&H_1^2 + H_2^2 + H_3^2 = 1 && \text{sphere of constant moment} \\
&\frac{H_1^2}{2J_1E_r} + \frac{H_2^2}{2J_2E_r} + \frac{H_3^2}{2J_3E_r} = 1 && \text{ellipsoid of constant energy}
\end{aligned}
$$

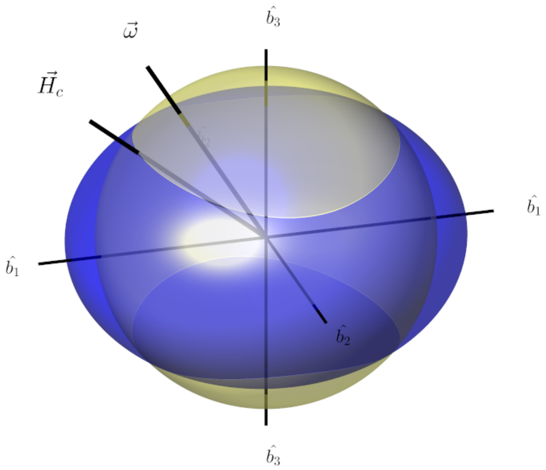


FIGURE 1. Sphere of constant moment and ellipsoid of constant energy

We can also draw the ellipsoids of constant moment and of constant energy and their intersection.

$$
\begin{aligned}
&\frac{J_1^2}{H_c^2}\omega_1^2 + \frac{J_2^2}{H_c^2}\omega_2^2 + \frac{J_3^2}{H_c^2}\omega_3^2 = 1 && \text{ellipsoid of constant moment} \\
&\frac{J_1}{2E_r}\omega_1^2 + \frac{J_2}{2E_r}\omega_2^2 + \frac{J_3}{2E_r}\omega_3^2 = 1 && \text{ellipsoid of constant energy}
\end{aligned}
$$

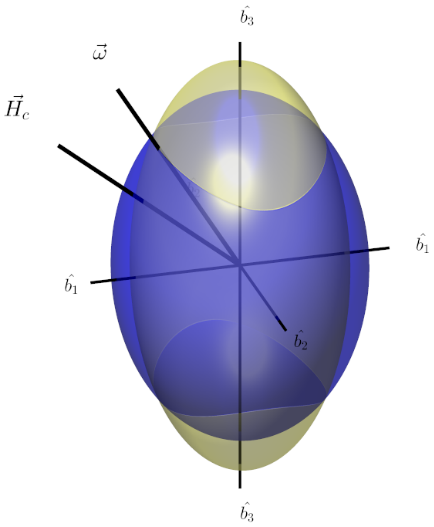


FIGURE 2. Ellipsoid of constant moment and ellipsoid of constant energy

2 Part 2

2.1 Problem

Draw the polhode (including the separator) for different kinetic energy values on the ellipsoid of constant moment.

2.2 Polhode representation

We need to create of vector of different values for the kinetic energy, ranging from the min to the max. We'll use these values to draw the corresponding polhodes on the ellipsoid of constant moment. We need to make sure the vector passes through E_2, as we want to draw the separator as well.
For each value of $\vec{E}_r$ we'll need to calculate the angular velocity vector $\vec{\omega}$ and the angular momentum vector $\vec{H}_c$. These vectors were already calculated before (for another value), so we'll gloss over the computations.

$$\vec{E}_r = [E_{min}, \dots, E_2, \dots, E_{max}]$$

$$E_r = \vec{E}_r(i)$$
$$\vec{\omega} = \begin{bmatrix} \omega_{10} & \omega_{20} & \omega_{30} \end{bmatrix}$$
$$\vec{H}_c = \begin{bmatrix} J_1\omega_{10} & J_2\omega_{20} & J_3\omega_{30} \end{bmatrix}$$

We then draw the ellipsoid of constant moment, which will be the same for all values of $\vec{E}_r$. Let's use the notation below (k_1, k_2 and k_3)to ease calculations.
Using the *Euler's rotation equations* we can estimate the path that $\vec{\omega}$ takes over time, and therefore the polhode for different values of kinetic energy. This is done through integration, since the 1st order differential system is a Cauchy problem. We also have to remember we're solving the Euler equations when there is no torque involved (so $M_1 = M_2 = M_3 = 0$, meaning free motion).

$$k_1 = \frac{J_2 - J_3}{J_1}$$

$$k_2 = \frac{J_3 - J_1}{J_2}$$
$$k_3 = \frac{J_1 - J_2}{J_3}$$

$$\begin{cases} \dot{\omega_1} = k_1 \omega_2 \omega_3 \\ \dot{\omega_1} = k_2 \omega_3 \omega_1 \\ \dot{\omega_1} = k_3 \omega_1 \omega_2 \end{cases} \qquad + \qquad \begin{cases} \omega_1(t=0) = \omega_{10} \\ \omega_2(t=0) = \omega_{20} \\ \omega_3(t=0) = \omega_{30} \end{cases}$$

Once the components of $\vec{\omega}(t)$ are found, it is possible to draw the polhode for different values of kinetic energy on the ellipsoid, as depicted in Fig. 3.

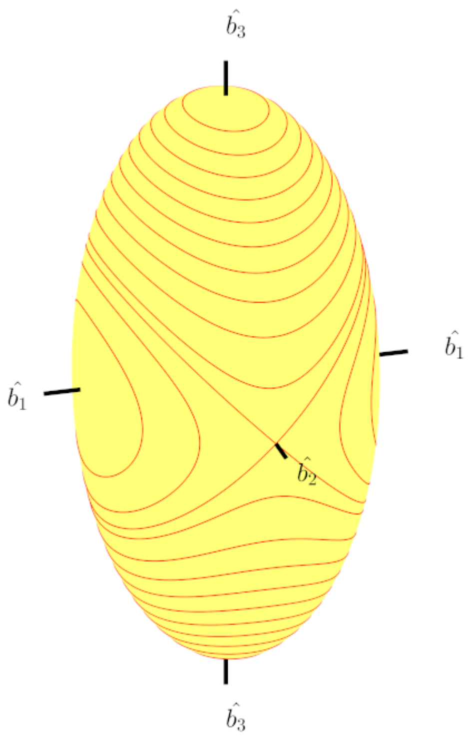


FIGURE 3. Polhode for different energy values

3 Part 3

3.1 Problem

Then choose a value for E_r and draw the polhode and the herpolhode with the plan normal to $\vec{H}_c$. Also draw the ellipsoid of constant energy that should roll on the plane.

3.2 The plane and the body attitude

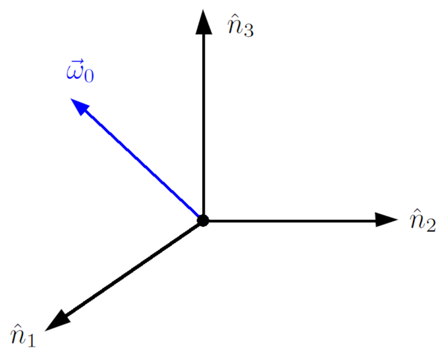
The vector $\vec{H}_c$ in the inertial reference frame is constant and fixed in place. We define the plane orthogonal to $\vec{H}_c$, distant d from the origin of the frame. The erpolhode exists only on that plane.

$$d = \frac{\vec{H}_c \cdot \vec{\omega}}{|\vec{H}_c|} = \frac{2E_r}{|\vec{H}_c|}$$

The initial attitude of the body must be estimated using the knowledge on $\vec{H}_c$ and $\vec{\omega}$. This allows us to calculate the attitude BN and at the end obtain it over a period of time.
We're able to obtain $\cos\alpha_{11}$, $\cos\alpha_{21}$, $\cos\alpha_{31}$, $\cos\alpha_{12}$, $\cos\alpha_{22}$ and $\cos\alpha_{33}$ from BN_0. Then, integrating the direction cosines and the components of the angular speed over time we get $BN(t)$ and $\vec{\omega}$.

$$\hat{n}_3 = \frac{\vec{H}_c}{|\vec{H}_c|}$$
$$\hat{n}_2 = \frac{\vec{H}_c \wedge \vec{\omega}}{|\vec{H}_c \wedge \vec{\omega}|}$$
$$\hat{n}_1 = \hat{n}_2 \wedge \hat{n}_3$$

$$BN_0 = \begin{bmatrix} n_1 & n_2 & n_3 \end{bmatrix}$$
$$\dot{BN} = -[\tilde{\omega}] \cdot BN$$



3.3 Polhode and erpolhode

The erpolhode is simply the trace left by the polhode on the ellipsoid of constant energy, which rolls without slipping on the plane previously defined. It is possible then to use the attitude matrix BN at every single instant of time to rotate $\vec{\omega}$ and obtain the components along the inertial frame, thus defining the erpolhode.

$$\vec{\omega} = \begin{bmatrix} \omega_1(t) \\ \omega_2(t) \\ \omega_3(t) \end{bmatrix} \Bigg|_{body\ frame}$$
$$\vec{\omega}|_{inertial} = BN^T\ \vec{\omega}|_{body\ frame}$$

$$\text{erpolhode} = \{BN^T(t)\ \omega_B(t)\}$$
$$\text{polhode} = \{BN^T(t^*)\ \omega_B(t)\}$$

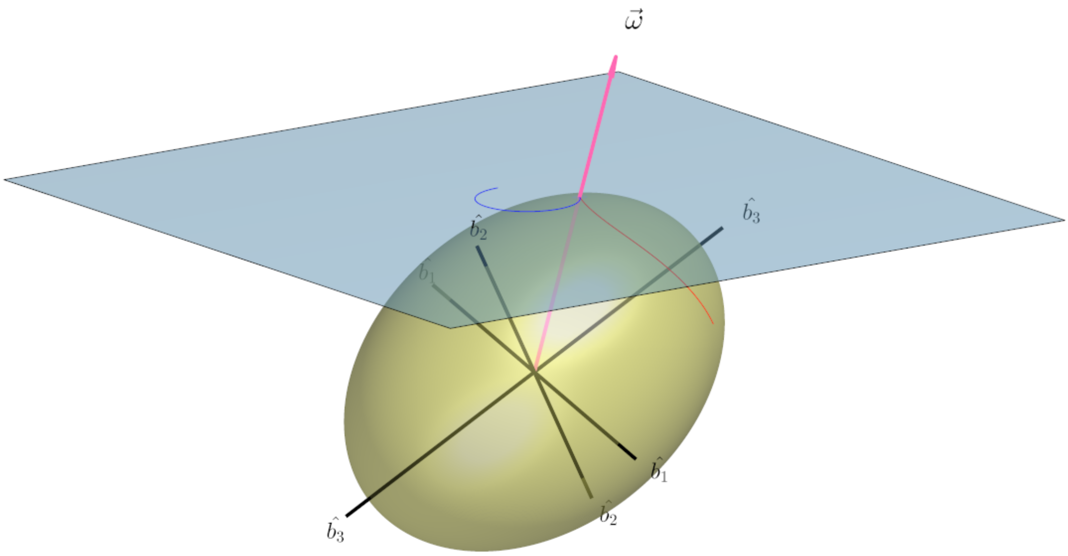


FIGURE 4. Polhode, herpolhode and ellipsoid of constant energy rolling on the plane

4 Part 4

4.1 Problem

Simulate the damping with the energy well model (i.e. applying a pair), reporting the $\vec{\omega}$ path on the ellipsoid using the recommended data and a second time by entering 0.45 for E_r. At the end of the integration, show the trends over time of E_r, $\vec{H}_c$, $\vec{\omega}$ (for the last two graph modules and components). The braking torque, orthogonal to $\vec{H}_c$ and in the plane ($\vec{H}_c$, $\vec{\omega}$), is chosen in order to achieve the law

$$E_r = E_r(0) - (E_r(0) - E_r(\infty))\exp(-\epsilon t)$$
$$\frac{dE_r}{dt} = \vec{T}_c \cdot \vec{\omega}$$

with $\epsilon = 5/t_{max}$ (in a realistic simulation should be a lot smaller)

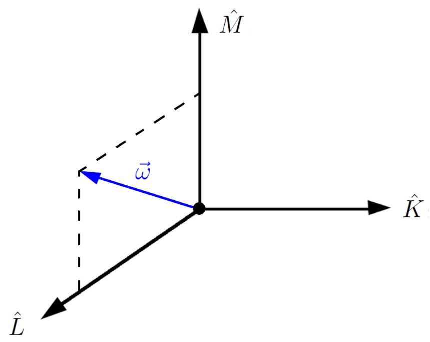
4.2 Energy well model

If we decide to visualize the dissipation of the kinetic energy over a time period we'll have to consider the appropriate dissipation coefficient.

$$\text{time} = 100\ s$$
$$\epsilon = \frac{5}{\text{time}} = 0.05$$

To simulate the dissipation through the energy well method it is necessary an proper torque, capable of affecting the body in a realistic manner. To do this, let's define a new reference frame, which will be used to construct this torque. We notice as the vector $\vec{\omega}$ is always inside the plane formed by $\hat{M}$ and $\hat{L}$, and by default orthogonal to the axis $\hat{K}$.

$$
\begin{aligned}
\hat{M} &= \frac{\vec{H}_c}{|\vec{H}_c|} \\
\hat{K} &= \frac{\vec{H}_c \wedge \vec{\omega}}{|\vec{H}_c \wedge \vec{\omega}|} \\
\hat{L} &= \hat{K} \wedge \hat{M}
\end{aligned}
$$



We do not want to modify the module of $\vec{H}_c$, that means the engineered torque $\vec{T}_c \perp \vec{H}_c$. If we hypothesize that the torque belongs to the plane created by $\vec{H}_c$ and $\vec{\omega}$ then $\vec{T}_c \cdot (\vec{H} \wedge \vec{\omega}) = 0$ is certainly true and the torque has its only component along the $\hat{L}$ axis. We still need to keep in mind that as the body rotates (and the vector $\vec{H}_c$ changes direction), so do the reference frame designed here. It will be necessary to construct it at every single time. We can estimate the module of $\vec{T}_c$ by simply inverting the given law of dissipation of energy.

$$
\begin{aligned}
E &= \frac{1}{2}\omega^T J\omega \\
\frac{dE}{dt} &= \omega^T (J\dot{\omega}) \\
&= \omega^T(-\omega \wedge J\omega + T_c) \\
&= \omega^T(-\omega \wedge J\omega) + \omega^T T_c \\
&= J\omega^T \cancel{(\omega \wedge \omega)} + \omega^T T_c
\end{aligned}
$$

$$
\Rightarrow \frac{dE}{dt} = \vec{T}_c \cdot \vec{\omega} = T_c \hat{L} \cdot \vec{\omega}
$$

$$
\vec{T}_c = \frac{\frac{dE}{dt}}{\hat{L} \cdot \vec{\omega}} \hat{L} = \frac{\epsilon(E_{min} - E_r)e^{-\epsilon t}}{\hat{L} \cdot \vec{\omega}} \hat{L}
$$

Lastly, by using the *Euler's equations* and integrating for the components of the angular speed, we're able to obtain the trends of the module and components of $\vec{\omega}$ over time. We can draw the polhode on the ellipsoid of constant moment, but this time it won't closed, but open as the kinetic energy of the body is slowly dissipating away, and the body goes from rotating around $\hat{b}_3$ to rotating around $\hat{b}_1$ (a stable rotation), the axis with highest moment of inertia J_1.

$$
\begin{aligned}
\dot{\omega}_1 &= k_1\omega_2\omega_3 + \frac{T_c(1)}{J_1} \\
\dot{\omega}_2 &= k_2\omega_1\omega_3 + \frac{T_c(2)}{J_2} \\
\dot{\omega}_3 &= k_3\omega_1\omega_2 + \frac{T_c(3)}{J_3}
\end{aligned}
$$

4.3 Figures

- Case a : suggested values from the beginning

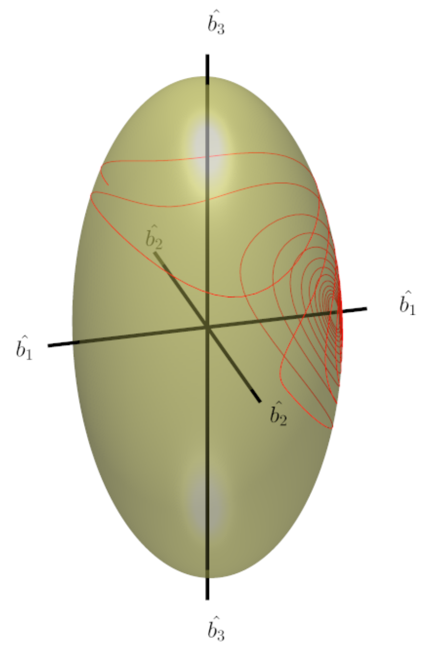


FIGURE 5. Effect of dissipation on the polhode (case a)

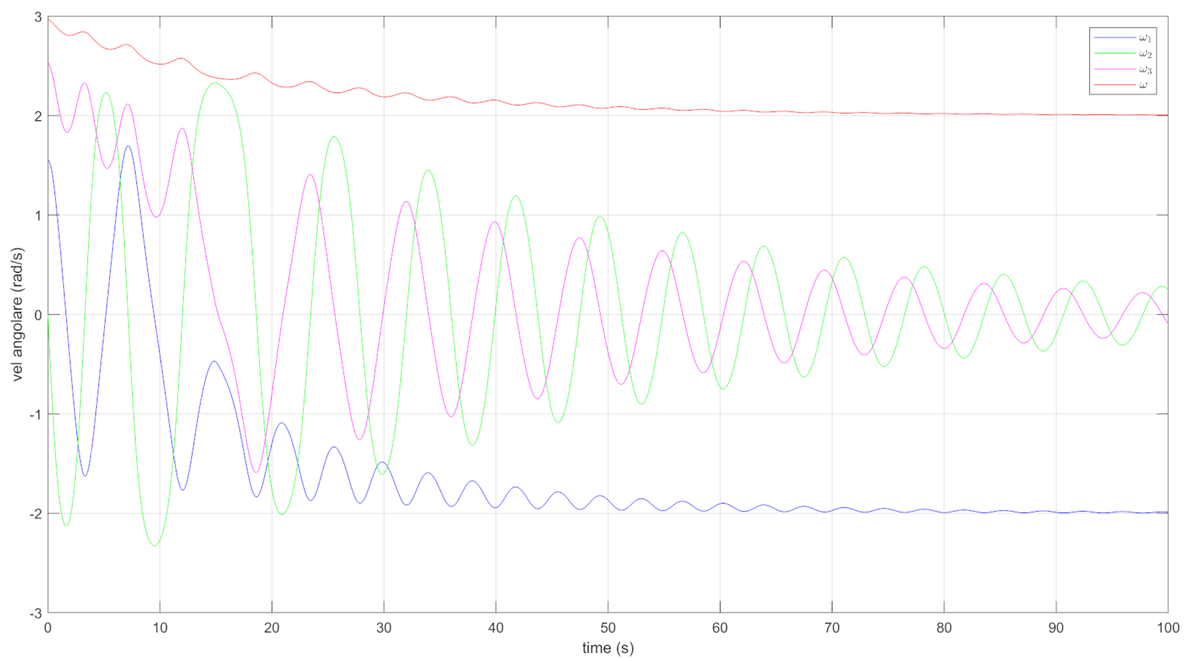


FIGURE 6. Effect of dissipation on $\vec{\omega}$ (case a)

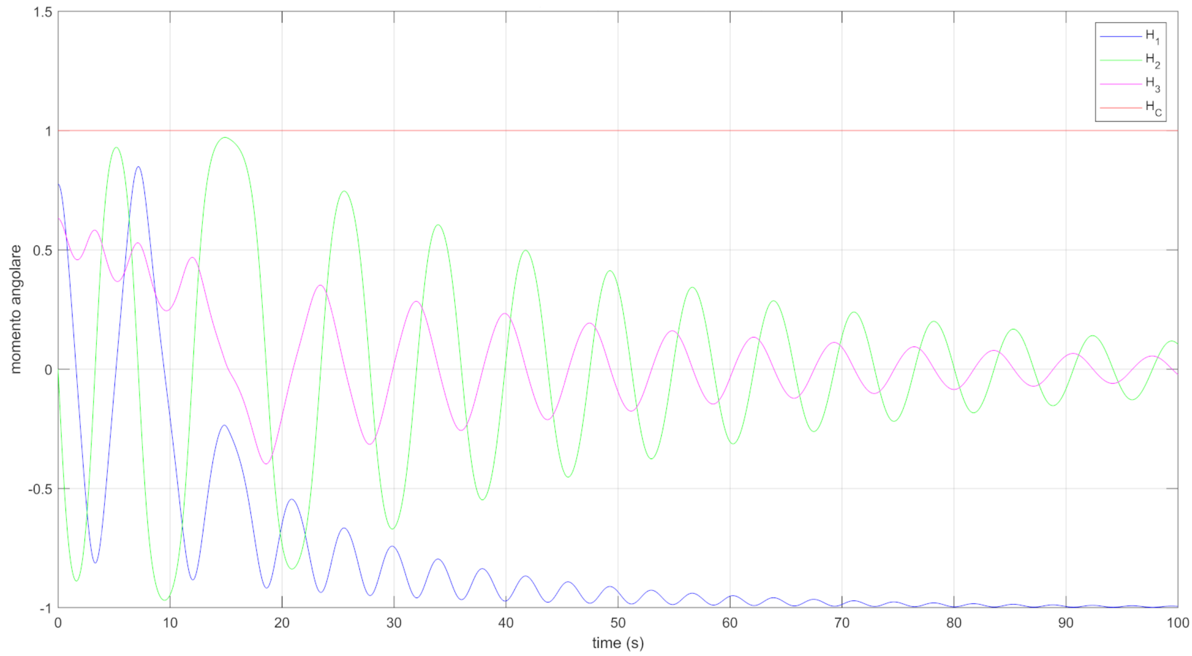


FIGURE 7. Effect of dissipation on $\vec{H}_c$ (case a)

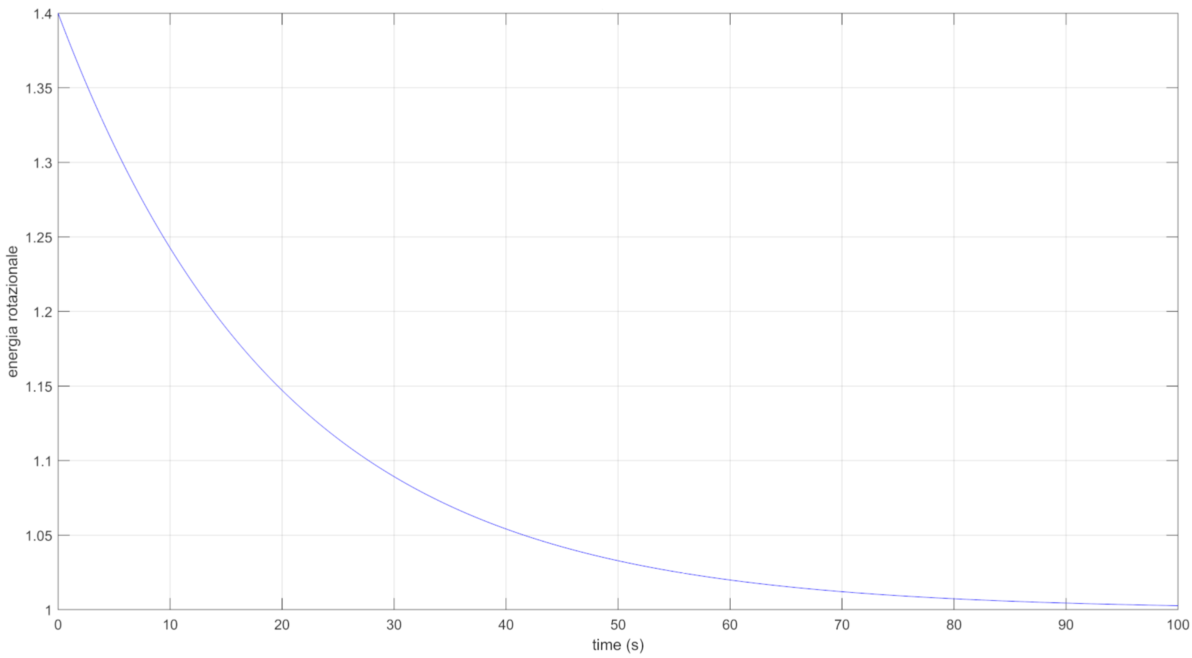


FIGURE 8. Effect of dissipation on E_r (case a)

- Case b : modified value for E_r

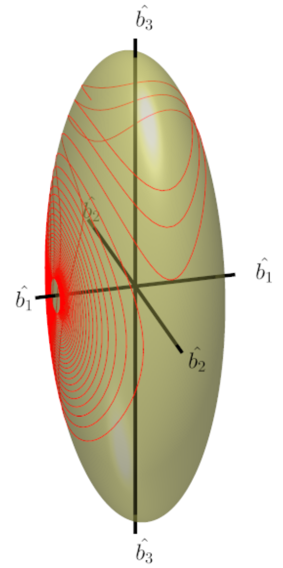


FIGURE 9. Effect of dissipation on the polhode (case b)

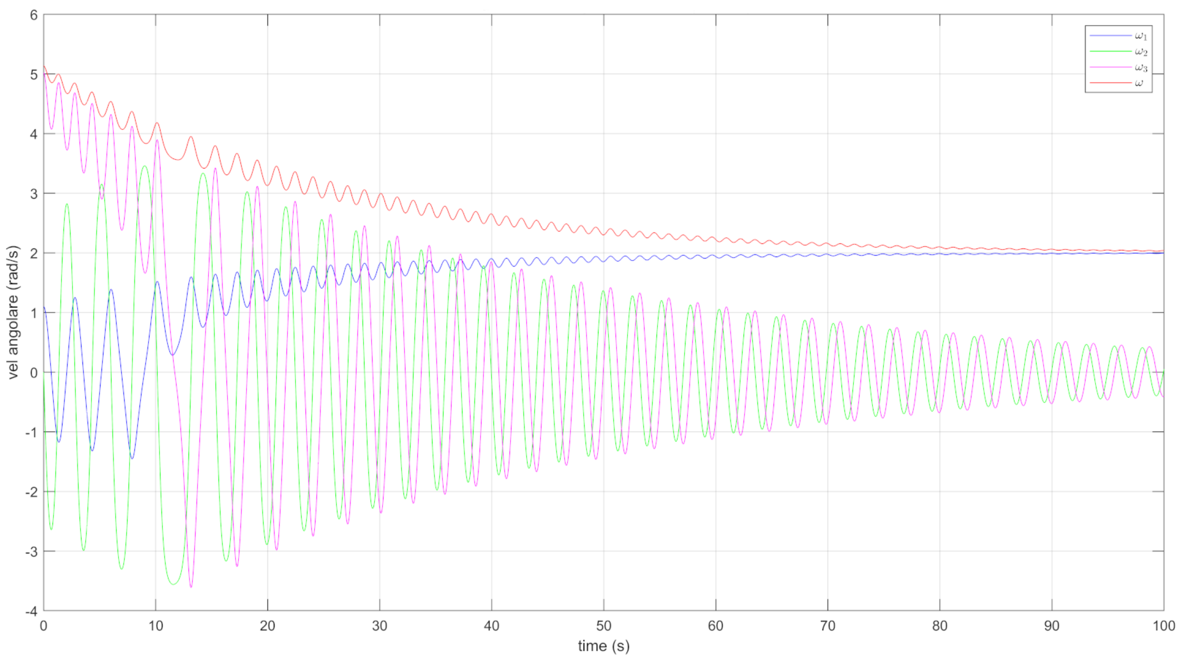


FIGURE 10. Effect of dissipation on $\vec{\omega}$ (case b)

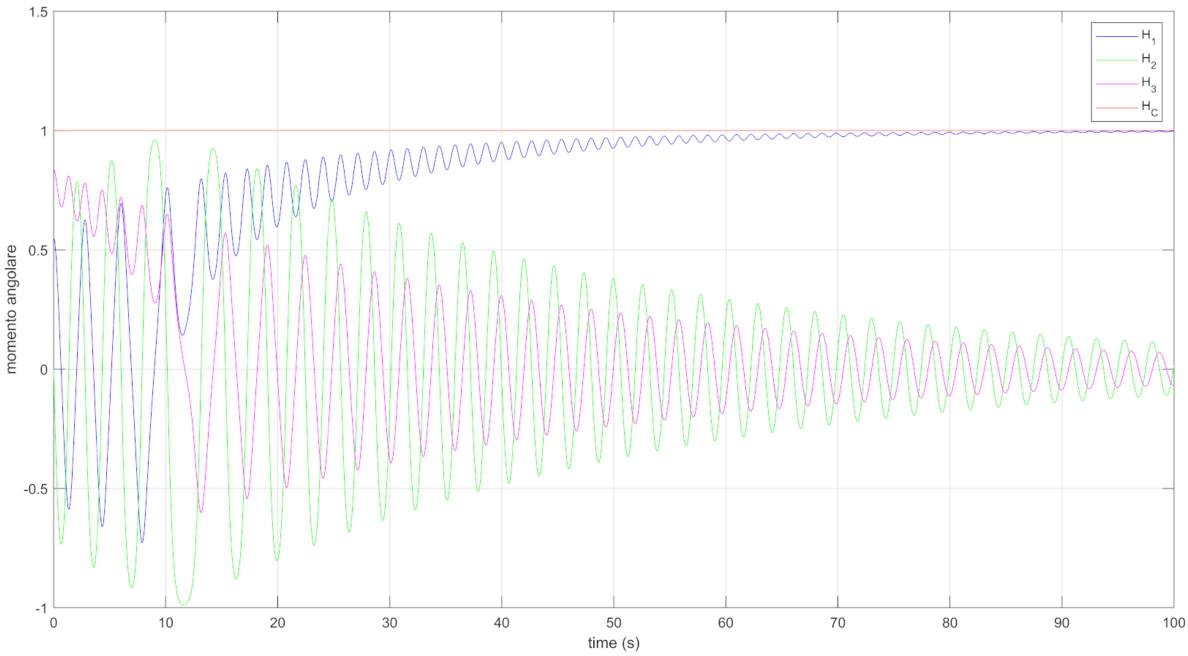


FIGURE 11. Effect of dissipation on $\vec{H}_c$ (case b)

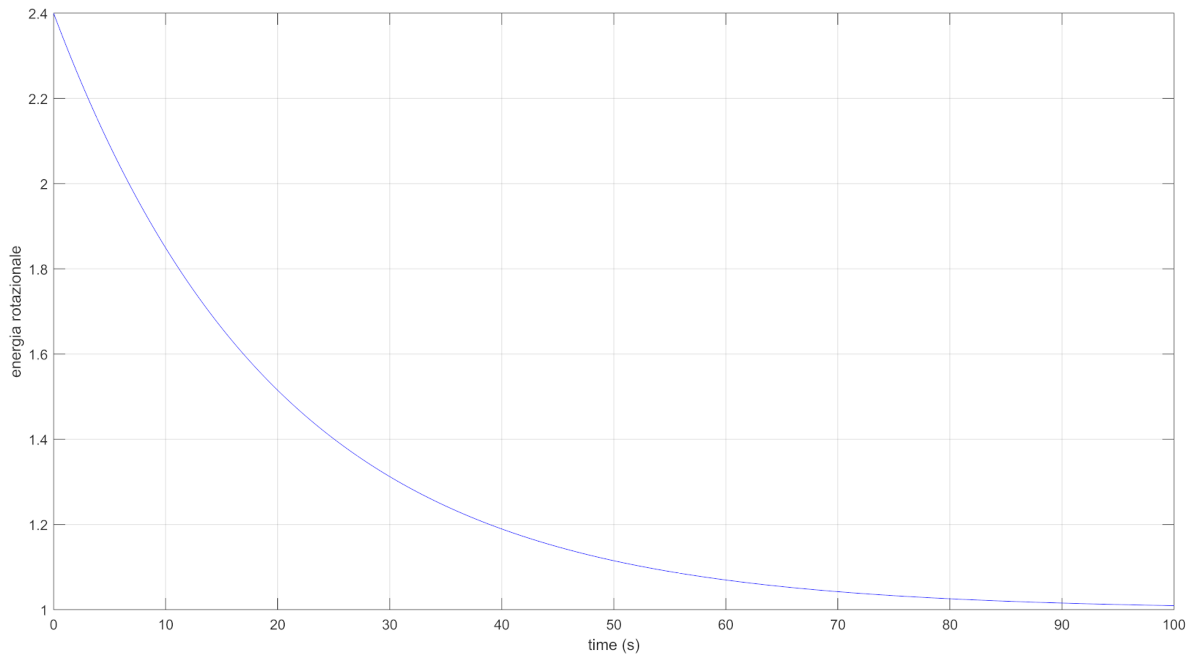


FIGURE 12. Effect of dissipation on E_r (case b)